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Growth & Development

Concept, Factors, Aging, Somatotypes



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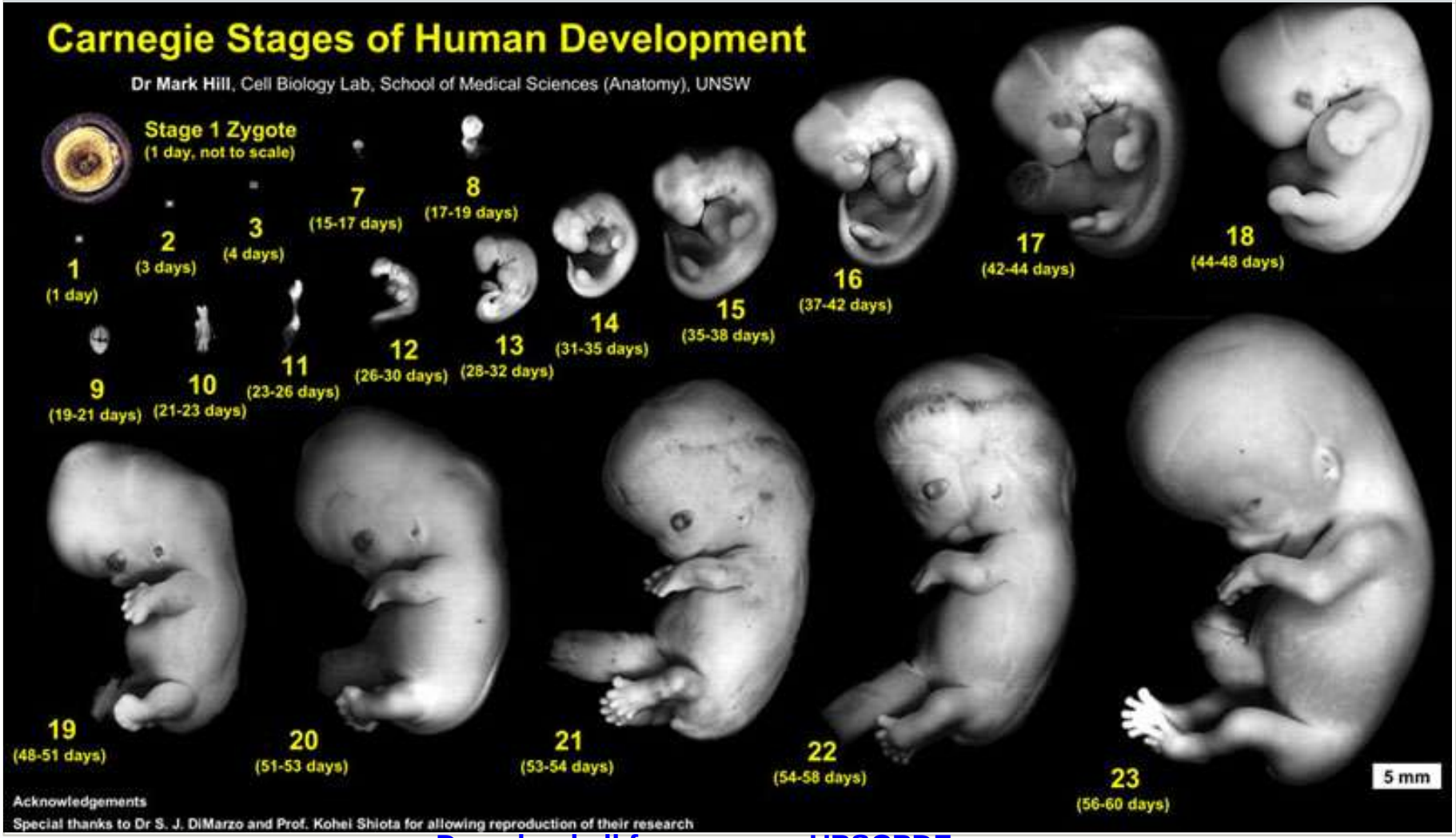
Concept of G & D

- Different dynamic aspects
- Both begin with fertilization
- Refer to quantitative and qualitative changes
- **Growth**
  - Increase in the number of cells
  - Maturation and Expansion
  - Increase in the size of the body
- **Development**
  - Functional specialization

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Prenatal G & D



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Aspects of G & D

- Physical Growth and Development
- Intellectual Development – Learning and Memory
- Social and Personality Development





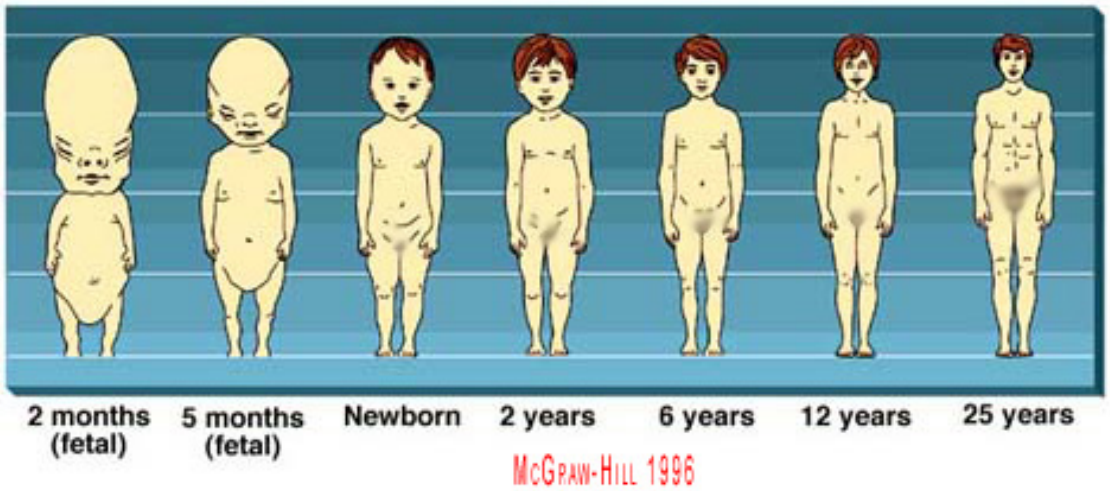


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Principles of G & D

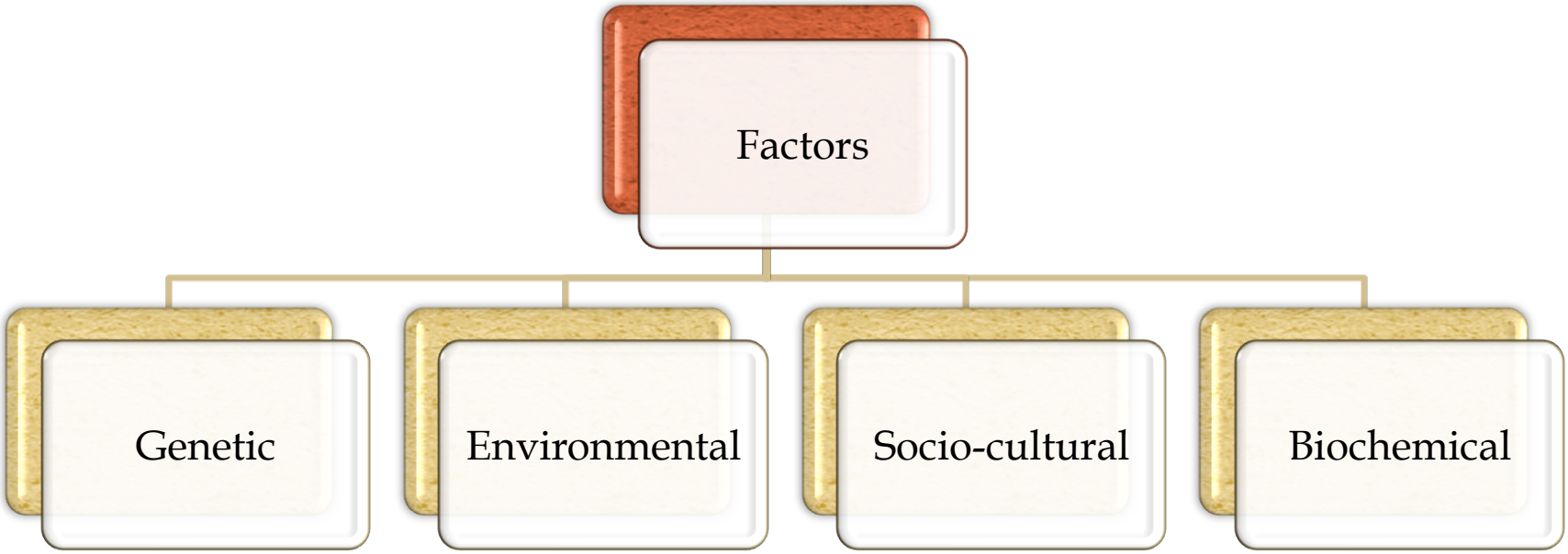
1. Cephalo-caudal Principle
2. Proximo-distal Principle
3. Simple to Complex



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Factors in G & D



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Genetic Factors

1. Genes control coordination and sequence of G and D
   - Inducers – House Keeping / Luxury Genes
   - Bio Rhythms
2. Cell Division
   - DNA replication
   - Synthesis of proteins
3. Metabolism
4. Homeotic Genes – specialization of organs
5. Heredity
   - Phenotype
   - Sex
   - Twins

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Environmental Factors

1. Migration – adaptive modifications
2. Climate – Bergman's and Allen's rules
3. Emotions – Social environment
   - Anxiety, insecurity, lack of emotions – affect hormones responsible for G and D
4. Vulnerable groups
   - Pregnancy
   - Learning
   - Homozygotes

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Socio-cultural Factors

1. Nutritional Deficiency
2. Home environment – emotions
3. Faulty family budgeting
4. Minor illnesses – malnutrition
5. Illiteracy
6. Habits – Alcoholism, drug addiction, smoking etc.

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Biochemical Factors

Prenatal	Postnatal	Adolescence
• Thyroxin (DNA & RNA Activation) • Chorionic Gonadotropin (Gonadal differentiation)	• Thyroxin • Growth Hormone (Cell division etc.)	• Thyroxin • Growth Hormone • Luteinizing Hormone (Ovulation & Testosterone) • Follicle Stimulating Hormone (Gametogenesis)

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Abnormal Growth

PRENATAL FACTORS

- **Intra Uterine Growth Retardation**
  - Maternal Malnutrition
  - Maternal Illness
  - Medications
  - Maternal habits like alcoholism (FAS), smoking etc.
- **Genetic Disorders**
  - Inborn errors of metabolism – genetic disorders
  - Turner's syndrome etc. – chromosomal abnormalities

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ABNORMAL GROWTH

POSTNATAL FACTORS

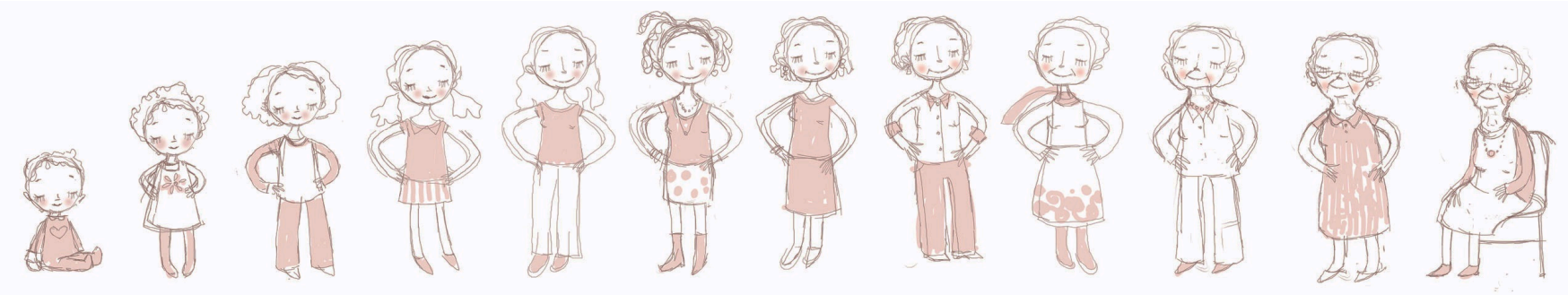
1. Nutritional Dwarfism
   - Protein / Zinc / Iron Deficiency
2. Chronic Visceral Diseases
   - Heart/lung/gastro – stunted growth
   - Chronic infections
3. Endocrine disorders
4. Emotional deprivation

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AGING

Process, Theories…



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Characteristics of Aging

1. Changes associated with aging increase the probability of death
2. Aging reduces the animal's capacity to cope with its environment
3. Changes associated with aging are cumulative – though death is sudden, the processes leading to death are gradual
4. Processes associated with aging and death are inescapable – intrinsic aspects of life

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Definitions

Aging

- ***Changes that are related to age, regardless of when they happen in life span***
- From conception to old age and death
- Biological Longevity

Senescence

- ***Changes occurring in the later part of life***
- Post-reproductive life
- Chronological Longevity

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Types of Aging

- **Primary Aging**
  - Process of bodily deterioration beginning at early stage and continues
  - Uncontrollable and natural process
- **Secondary Aging**
  - Result of abuse, disuse and disease
  - Controllable

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Theories of Aging

- **Cellular Theories** – intracellular errors & processes

1. Theory of Hayflick Limit – Leonard Hayflick
2. Theory of Free Radical Reactions – Lester Packer & James Smith
3. Theory of DNA Error – Ronald Hart & Richard Setlow
4. Error Catastrophe Theory – Robin Holliday
5. Theory of Induction / Missing Enzyme – M.S. Kanungo

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Theories of Aging

- **Extracellular Theories –** errors & processes outside cells

1. Immunological Theory – Burnet & Walford
2. Collagen Theory
3. Wear and Tear Theory

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Growth Study Methodologies

Cross Sectional, Longitudinal & Cross Sequential

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Methodologies

- Information about growth and development is most commonly gathered by

1. *Cross sectional studies*
2. *Longitudinal studies*
3. *Cross - Sequential designs*

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Cross Sectional & Longitudinal Studies

- In a **Cross Sectional Study,** people of different ages are assessed on one occasion.
- Cross sectional studies provide information about ***differences*** in development among different age groups
- In a **Longitudinal Study**, researchers measure the same people more than once to see changes in development over time – ***changes*** within same person over time.
- The researchers may measure one characteristic, such as vocabulary size, IQ, height or several factors at a time.
- Longitudinal methods study *process* of development

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Cross Sectional & Longitudinal Studies

- Cross-sectional studies, then, look at **differences** among groups of people; longitudinal studies assess **changes** undergone by one or more persons over a period of time.
- **Strengths of Cross sectional study**
  - Speed & economy
  - Does not loose subjects who drop out
- **Weaknesses of Cross sectional study**
  - Masking of differences between individuals as emphasis on group averages
  - It cannot eliminate cohort or generational influence on subjects born at different times

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Cross Sectional & Longitudinal Studies

- Great strength of *longitudinal* studies is their sensitivity to individual patterns of change
- Weakness – Expensive and time consuming
- Major weakness – bias because people who tend to volunteer are better socioeconomically
- Practice effect – subjects end up doing better in later tests

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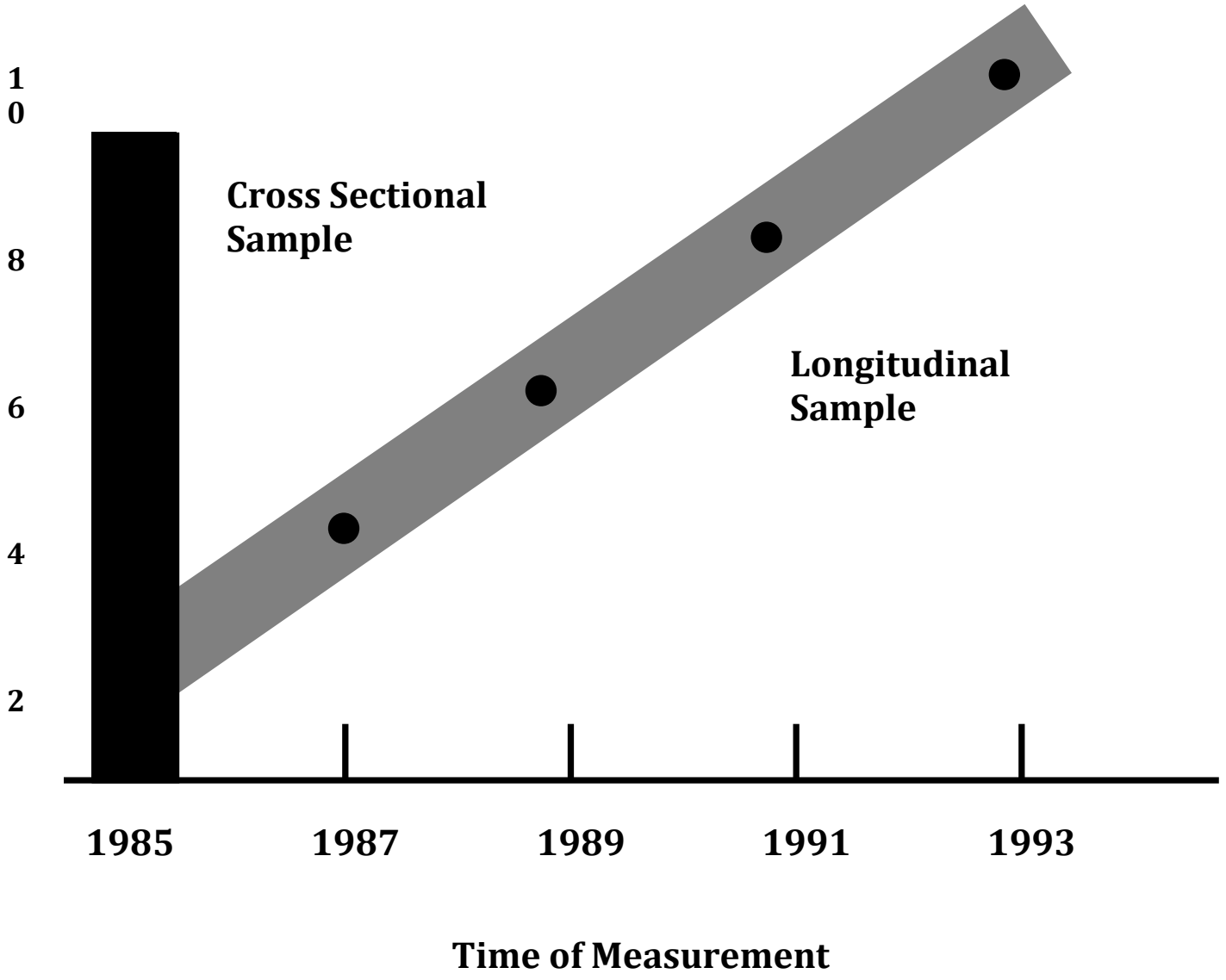
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Sequential Studies

- The **cross-sequential study** is one of several strategies designed to overcome the drawbacks of longitudinal and cross-sectional studies.
- This method combines the other two: people in a cross-sectional sample are tested more than once, and the results are analyzed to determine the differences that show up over time for the different groups of subjects.
- Main research though this study – Intellectual functioning – findings are more realistic

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